Module 10: Linux Exploitation and Privilege Escalation

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# Exercise 01: Performing Reconnaissance and Vulnerability Assessment using AI
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Scenario

AI-Assisted Reconnaissance & Automated Vulnerability Assessment leverages LLM-driven natural language shell interfaces—specifically **ShellGPT (`sgpt`)** powered by OpenAI GPT-4—to construct, refine, and execute complex security scanning pipelines directly from the command line interface.

Key technical concepts in ShellGPT AI reconnaissance:
1. **API Key Provisioning & Script Configuration**: Initializing ShellGPT environment using **bash sgpt.sh** and injecting ASPEN/iClass platform activation keys into local configuration stores (`~/.config/shell_gpt/.sgptrc`).
2. **Natural Language Shell Automation (`sgpt --shell`)**: Using the `--shell` flag to translate plain-English security intent into multi-step terminal command sequences.
3. **Automated Nmap Aggressive & Vulnerability Scanning**: Prompting ShellGPT to execute aggressive OS/service detection (`nmap -A`) and Nmap Scripting Engine vulnerability auditing (`nmap --script vuln`) against target Linux server **192.168.0.51**.
4. **Structured Report Generation**: Instructing ShellGPT to parse, format, and export structured reconnaissance output into a text file (**scan_results.txt**) for subsequent privilege escalation phase analysis.

In this exercise, we log into **Parrot Security** (**pentester** / **toor**), configure ShellGPT via `bash sgpt.sh`, submit a natural language prompt to perform aggressive and vulnerability scanning against target host **192.168.0.51**, execute the generated command pipeline (`E`), inspect the output file **scan_results.txt**, and identify the listening VSFTPD version (**2.3.4**).

Objective

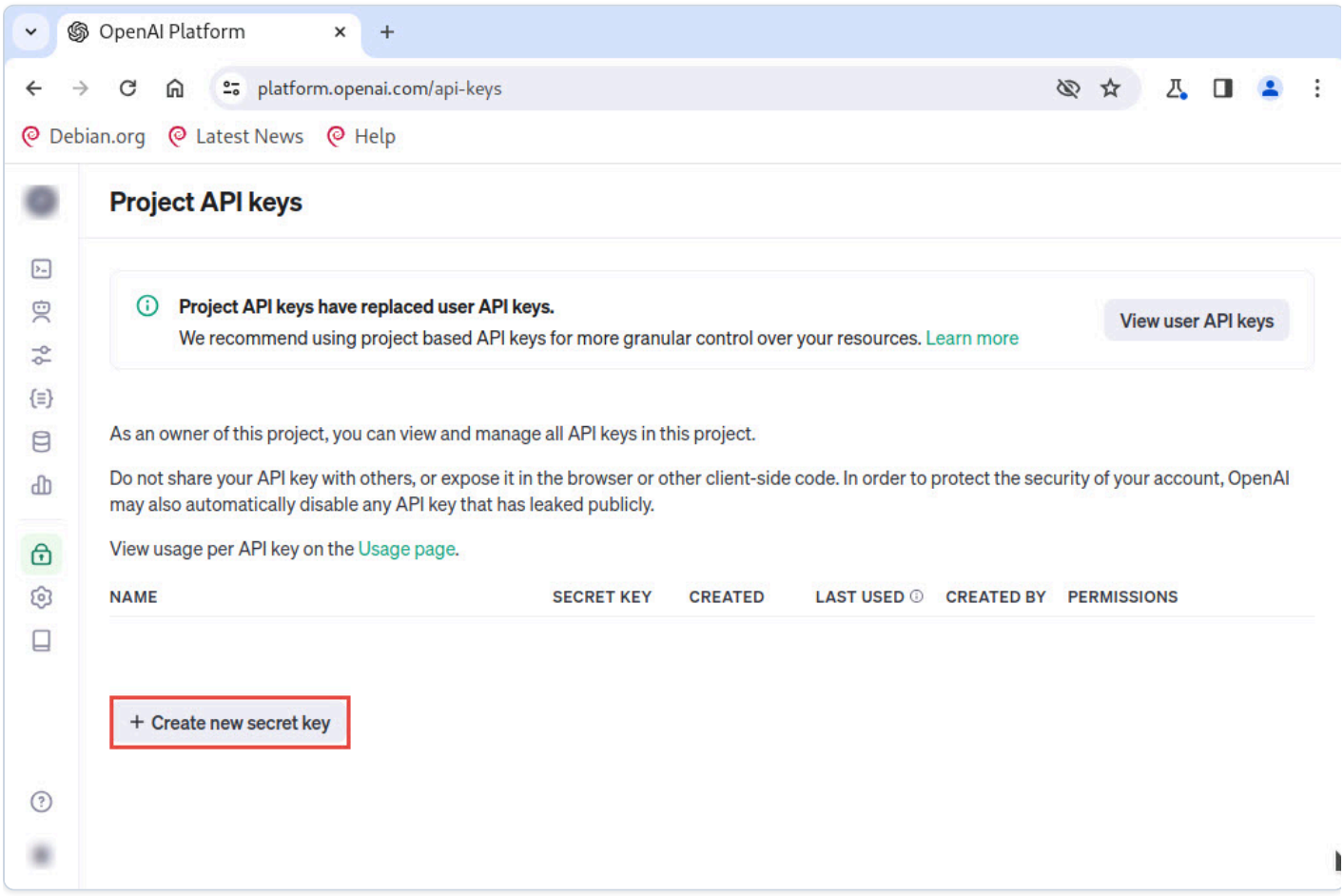
The objective of this lab is to help students learn how to:
- Authenticate to Parrot Security target VM (**pentester** / **toor**)
- Configure ShellGPT environment with AI activation key: **bash sgpt.sh**
- Generate automated Nmap scanning commands using natural language:
sgpt --shell "On the target IP address 192.168.0.51 use nmap to perform aggressive scan to identify OS and services and use nmap vuln script to perform vulnerability scan. Organize and display the results in clearly labeled sections for each type of information and save the results in a text file. Install all the required dependencies and libraries."
- Execute generated shell pipeline (`E`)
- Inspect reconnaissance output file: **cat scan_results.txt**
- Identify target FTP service version (**vsftpd 2.3.4**) on target machine **192.168.0.51**

Task 1: Configure ShellGPT, Execute AI-Driven Nmap Reconnaissance, and Analyze FTP Vulnerability Results

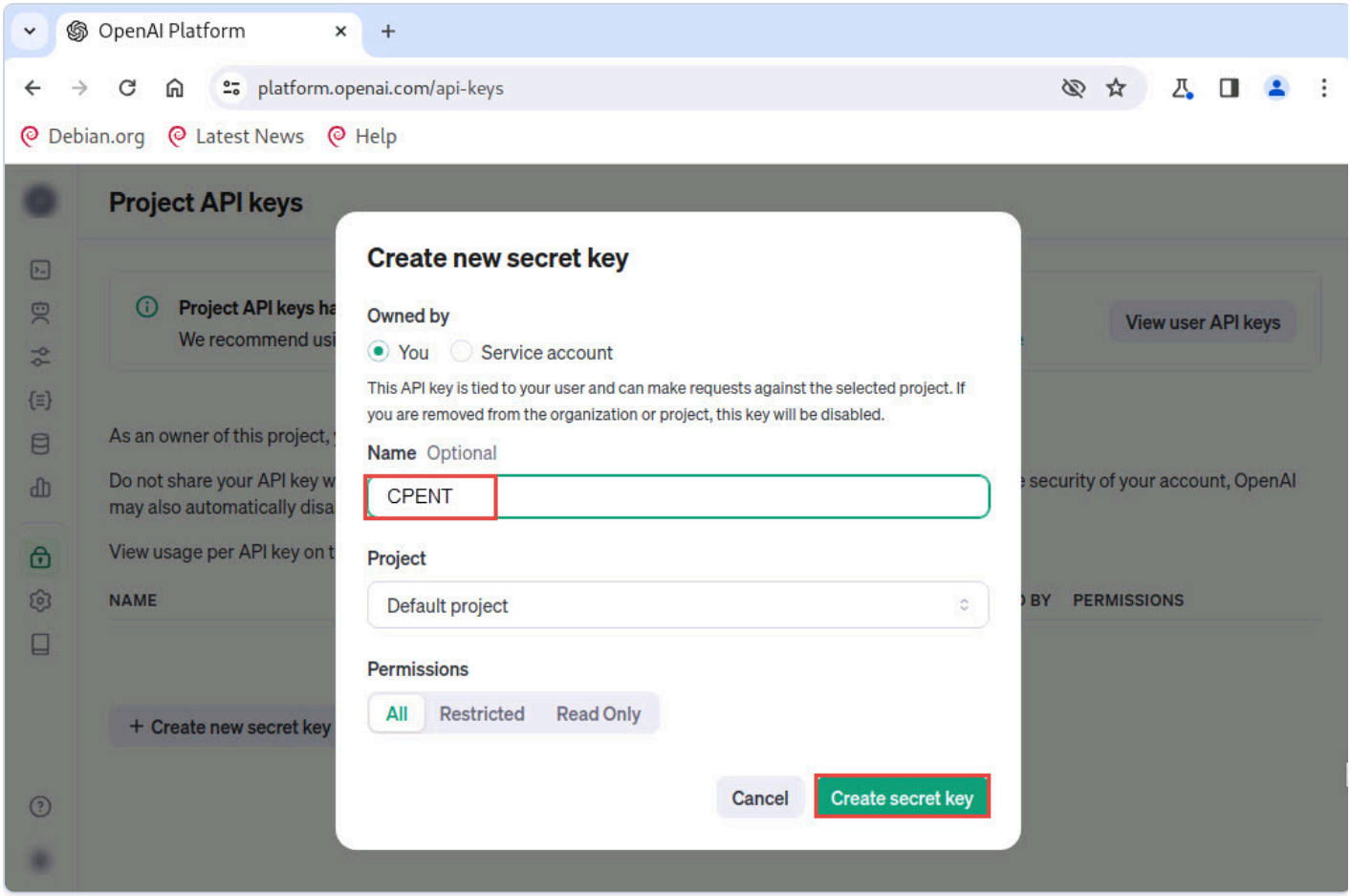
In this task, we configure ShellGPT with an API key, run a natural language command to initiate Nmap aggressive and vulnerability scans against `192.168.0.51`, save output to `scan_results.txt`, and audit FTP service version `2.3.4`.

Lab Tasks

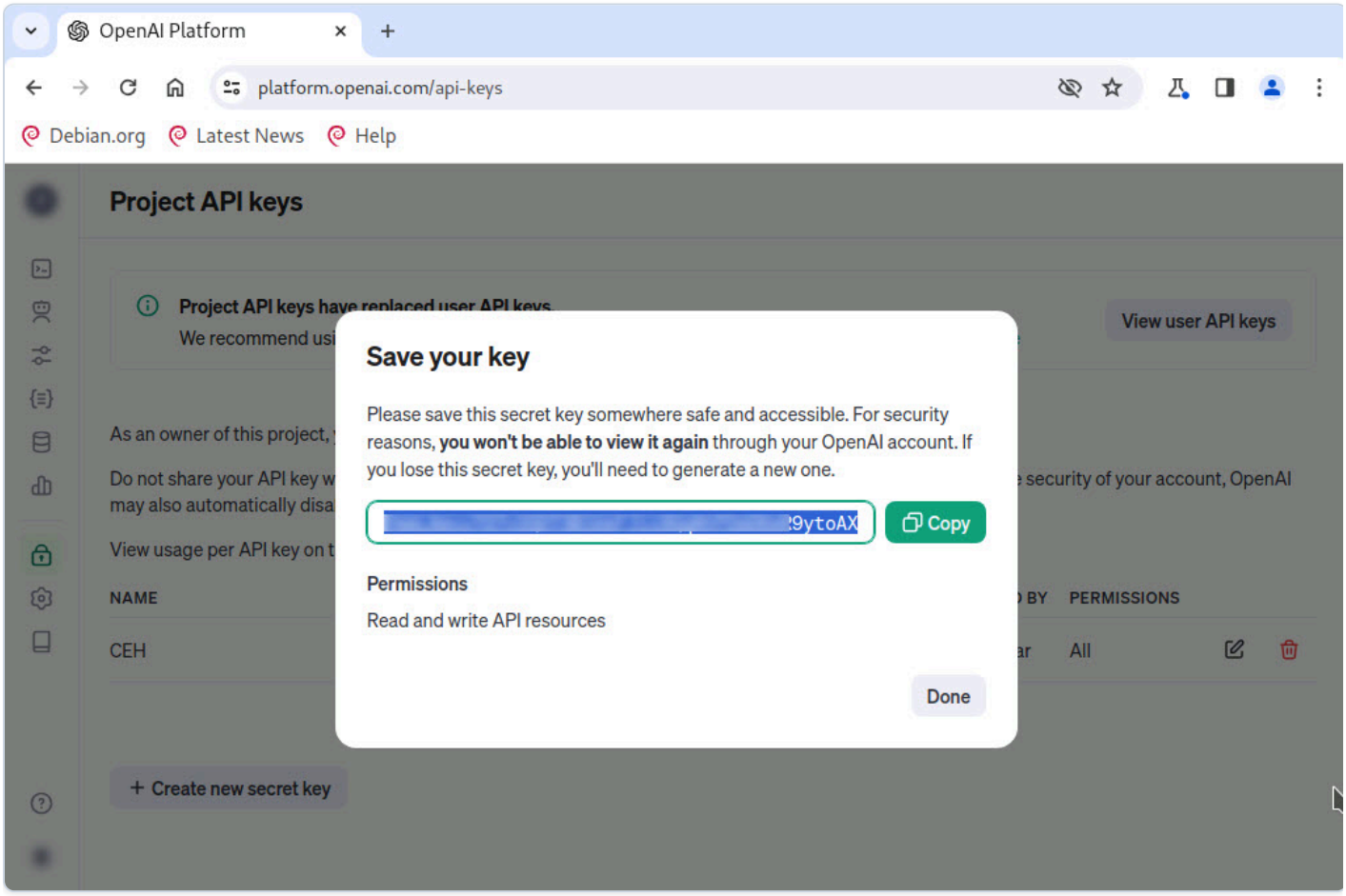
1. Log into **Parrot Security** VM using credentials (**pentester** / **toor**). Open terminal and obtain root privileges:
   sudo su



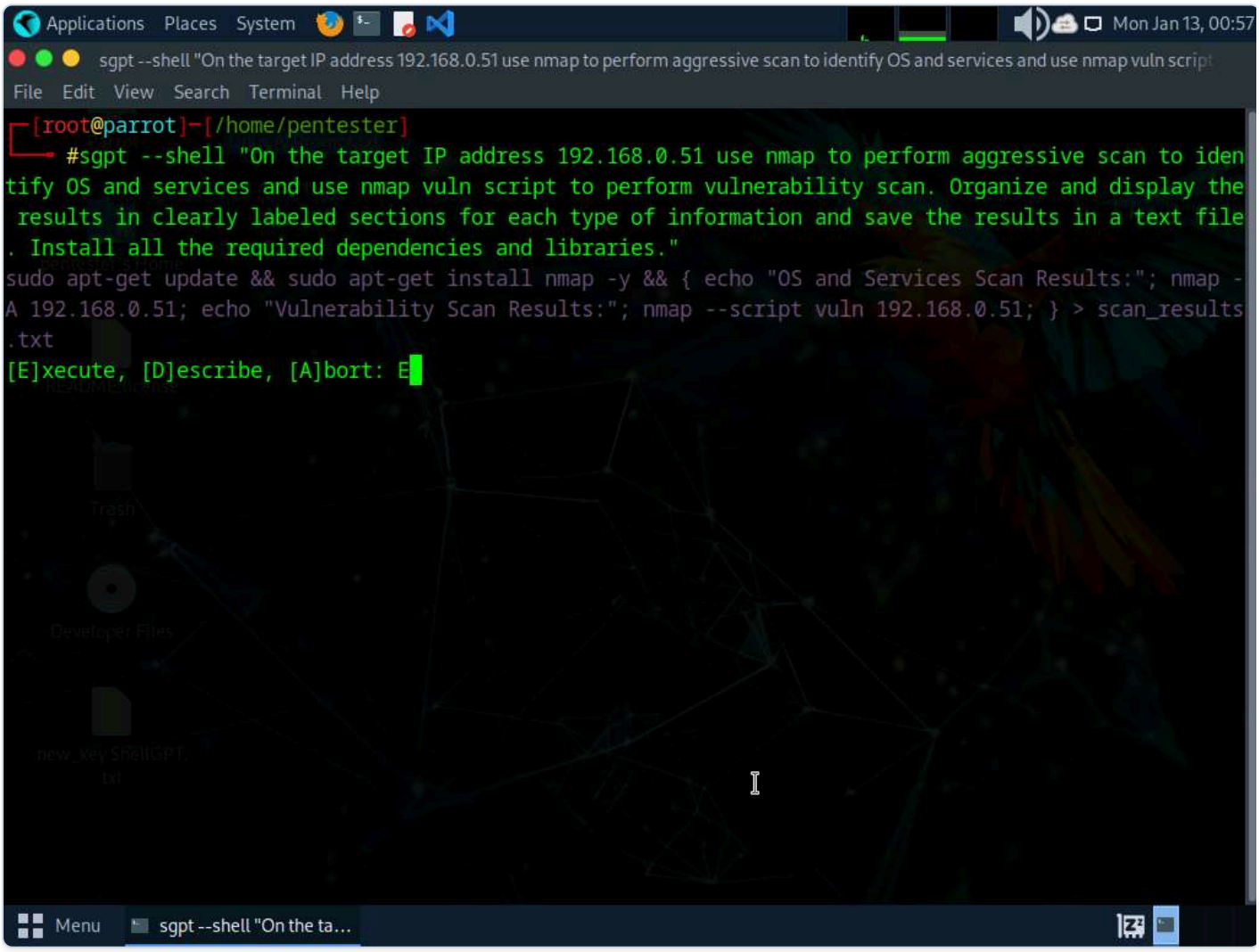
1. Configure ShellGPT by running setup script:
   bash sgpt.sh



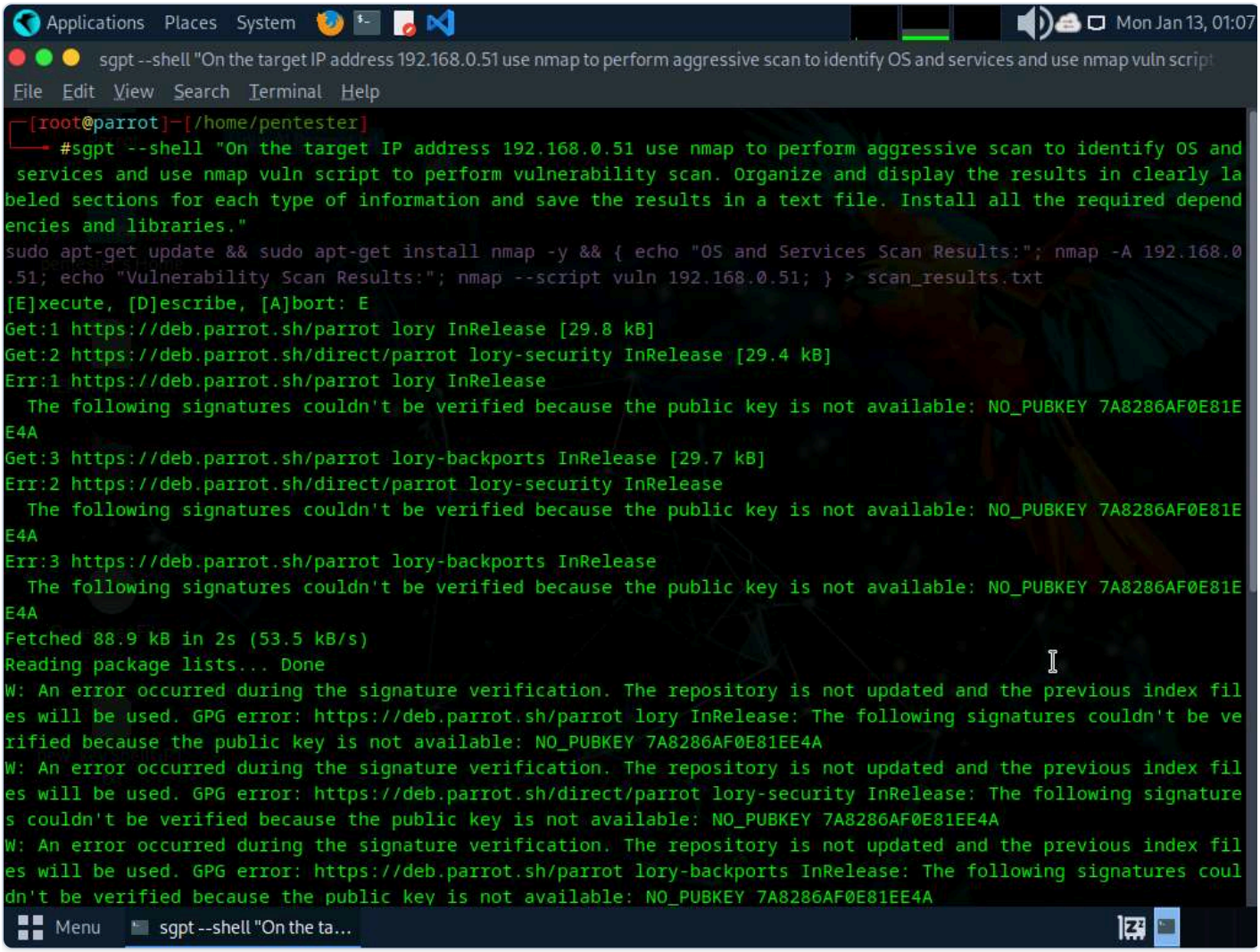
1. When prompted, enter your AI activation key. Verify success message upon completion.



1. Submit ShellGPT natural language prompt to generate and execute Nmap aggressive/vuln scanning commands against **192.168.0.51**:
   sgpt --shell "On the target IP address 192.168.0.51 use nmap to perform aggressive scan to identify OS and services and use nmap vuln script to perform vulnerability scan. Organize and display the results in clearly labeled sections for each type of information and save the results in a text file. Install all the required dependencies and libraries."



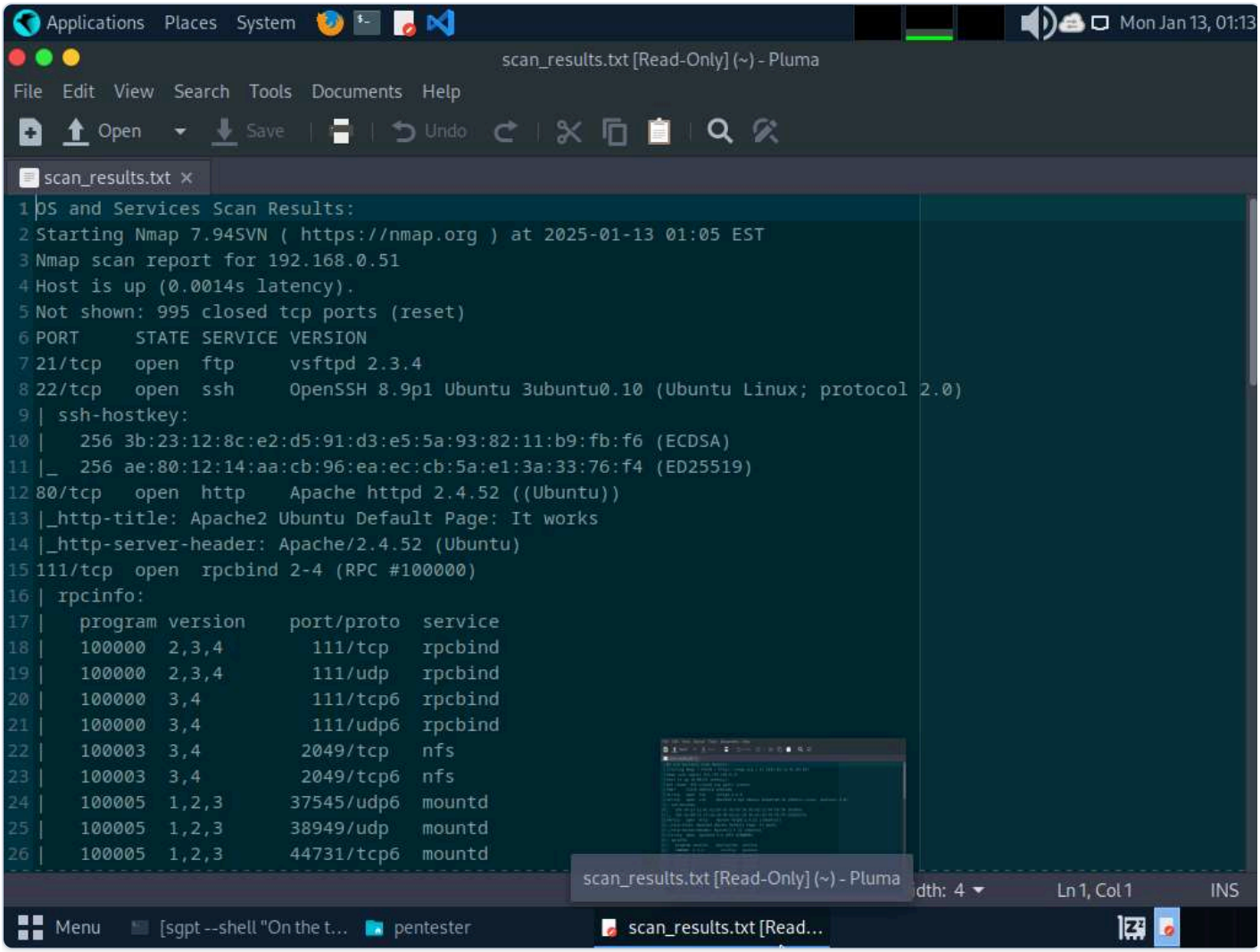
1. When prompted by ShellGPT, type **E** and press Enter to execute the generated scanning pipeline.

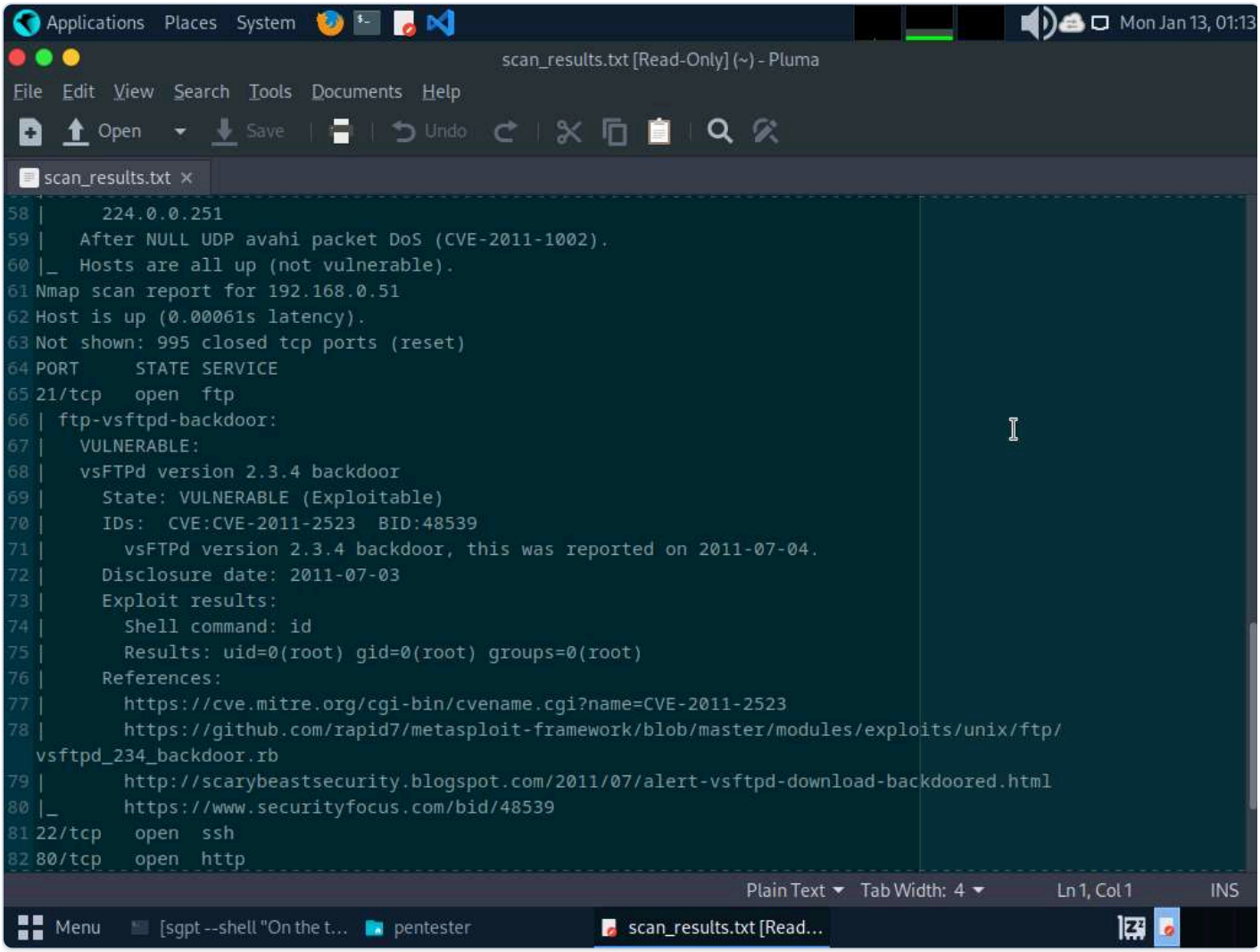


1. Open and view generated scanning output file (**scan_results.txt**):

   cat scan_results.txt

   Observe output: Identifies target OS, running services, and VSFTPD version **2.3.4** on TCP port `21`.


Applications Places System
Mon Jan 13, 01:13
scan_results.txt [Read-Only] (~) - Pluma
File Edit View Search Tools Documents Help
Open Save Undo
scan_results.txt
OS and Services Scan Results:
Starting Nmap 7.94SVN (https://nmap.org) at 2025-01-13 01:05 EST
Nmap scan report for 192.168.0.51
Host is up (0.0014s latency).
Not shown: 995 closed tcp ports (reset)
PORT STATE SERVICE VERSION
21/tcp open ftp vsftpd 2.3.4
22/tcp open ssh OpenSSH 8.9p1 Ubuntu 3ubuntu0.10 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
| 256 3b:23:12:8c:e2:d5:91:d3:e5:5a:93:82:11:b9:fb:f6 (ECDSA)
|_ 256 ae:80:12:14:aa:cb:96:ea:ec:cb:5a:e1:3a:33:76:f4 (ED25519)
80/tcp open http Apache httpd 2.4.52 ((Ubuntu))
|_http-title: Apache2 Ubuntu Default Page: It works
|_http-server-header: Apache/2.4.52 (Ubuntu)
111/tcp open rpcbind 2-4 (RPC #100000)
| rpcinfo:
| program version port/proto service
| 100000 2,3,4 111/tcp rpcbind
| 100000 2,3,4 111/udp rpcbind
| 100000 3,4 111/tcp6 rpcbind
| 100000 3,4 111/udp6 rpcbind
| 100003 3,4 2049/tcp nfs
| 100003 3,4 2049/tcp6 nfs
| 100005 1,2,3 37545/udp6 mountd
| 100005 1,2,3 38949/udp mountd
| 100005 1,2,3 44731/tcp6 mountd
scan_results.txt [Read-Only] (~) - Pluma
dth: 4
Ln 1, Col 1
INS
Menu
[sgpt --shell "On the t...
pentester
scan_results.txt [Read...



This concludes the lab exercise.

> **IMPORTANT:** ❓ **Question 10.1.1.1** On a Parrot Security machine, utilize Shell-GPT to determine the operating system and running services. Conduct a vulnerability scan on the target system (192.168.0.51) using the Nmap vulnerability script, and save the results in a .txt file. Additionally, identify the installed FTP version on the 192.168.0.51 machine. [!SUCCESS] ✅ **Answer:** **2.3.4**